Pseudocode — Previous Year Questions

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Q1 — fun1(p, q) recursive function (p=3, q=4) — Problem

- int fun1(int p, int q)
- if(q == 0) return 0
- if(q mod 2 == 0) return fun1(p+p, q/2)
- return fun1(p+p, q/2) + p
- End
- Find output for p=3, q=4. (Options: 7, 12, 8, None)

Recursive func -> calls itself -> loop

int fun1(int p, int q) // p=3, q=4; p=6, q=2; p=12, q=1

- if(q == 0) // p=24, q=0

  return 0

- if(q % 2 == 0) //even check
  - return fun1(p+p, q/2) //fun1(3+3, 4/2) = fun1(6, 2),

    // fun1(6+6, 2/2) ; fun1(12, 1)
- else
  - return fun1(p+p, q/2) + p // fun1(24, ½) + 12 → 0+12 =

    // 12(ans)

// ½ = 0.5 = (int)0.5 = 0;

Q1 — fun1(p, q) recursive function (p=3, q=4) — Step-by-step explanation

- Start: p=3, q=4.
  - q != 0 and q is even → call fun1(6,2).
  - q=2 even → call fun1(12,1).
  - q=1 odd → evaluate return fun1(24,0) + 12.
  - fun1(24,0) returns 0 because q==0 → final result = 0 + 12 = 12.
  - Answer: 12 (Option C).

Q2 — Logical AND/OR with integers — Problem

- x=2, y=1, z=5
- a = (x AND y) OR (z + 1)
- Print a
- (Options: 5, 3, 2, 1)

- x=2, y=1, z=5
- a = (x AND y) OR (z + 1)
- Print a

Solution

0 → false, else +ve all true

X AND Y -> 2 AND 1 -> T AND T → T

Z+1 → 5+1 = 6 -> T

T OR T -> T

F AND F → F

F AND T → F

T AND T → T

T AND F → F

F OR F → F

F OR T → T

T OR T → T

T OR F → T

Q2 — Logical AND/OR with integers — Step-by-step explanation

- Treat AND/OR as logical operations: (2 AND 1) is true → value 1.
  - (z + 1) = 6, so expression is 1 OR 6 → logical OR gives true (1).
  - Print 1.
  - Answer: 1 (Option D).

Q3 — Recursive sum function (a=2, b=3) — Problem

- doSomething(a,b):
- if(b == 1)
  - return 0
- else
  - return a + doSomething(a, b-1)
- Find output for a=2, b=3 (Options: 3,4,2,1)

Q3 — Recursive sum function (a=2, b=3) — Step-by-step explanation

- doSomething(2,3) = 2 + doSomething(2,2).
  - doSomething(2,2) = 2 + doSomething(2,1).
  - doSomething(2,1) returns 0.
  - So doSomething(2,2)=2, doSomething(2,3)=2+2=4.
  - Answer: 4 (Option B).

Fun(2,3) -> 2 + fun(2,2) → 2+(2 + fun(2,1)) → 2 + 2 + 0

Q4 — Digit counting and digit-sum for a=123 — Problem

- a=123, n=0;
- set q=a;
- while(q % 10 != 0) {
  - n = n+1;
  - q = q/10
- }
- print n
- Then while(a>0) {
  - r = a%10;
  - sum+=r;
  - a = a/power(10,n-1)
- }
- print sum
- (Options show pairs)

Q4 — Digit counting and digit-sum for a=123 — Step-by-step explanation

- First loop: q moves 123→12→1→0, so n increments 3 times → n=3 (printed).
  - Second loop: with n=3, a/power(10,2) is integer division by 100.
  - Iteration 1: r = 123%10 = 3 → sum=3; a = 123/100 = 1.
  - Iteration 2: r = 1%10 = 1 → sum = 4; a = 1/100 = 0.
  - Print sum = 4.
  - Answer: Prints '3' then '4' (Option D).

Q5 — If / Else-if chain (a=5, b=8) — Problem

- a=5, b=8
- if(a>8 OR b<7) c=a+b
- elseif(a+b>9) c=a-b
- else a=a+5; c=c-2
- a=a+b; print a and c

Q5 — If / Else-if chain (a=5, b=8) — Step-by-step explanation

- Check a>8 or b<7 → false.
  - Else-if a+b>9 → 13>9 true → c = a - b = 5 - 8 = -3.
  - After if: a = a + b = 13. Print a, c → 13, -3.
  - Answer: 13 -3 (Option D).

Q6 — While loop with assignment inside condition (infinite loop) — Problem

A. Code executes successfully and value of salary is displayed once.

B. Code executes successfully and nothing is displayed.

C. Code executes successfully and value of salary is displayed infinite number of times.

D. Compile time error.

```
1 Set Integer Emp_no=101
2 Set Integer salary=0
3 while(Emp_no=501)
4          salary=salary+100
5           display salary
6 end-while
```

Q6 — While loop with assignment inside condition (infinite loop) — Step-by-step explanation

- Using assignment inside a loop condition sets a value each check and typically yields a truthy value.
  - That keeps the loop condition true forever → infinite loop.
  - The salary value keeps increasing and is printed repeatedly.
  - Answer: Code runs infinitely; salary printed infinite times (Option C).

Q7 — do/while with alternating increment/decrement — Problem

A. Code will run infinite number of times.

B. The program will not enter the loop.

C. Code will execute and value of res will be displayed twice.

D. Code will execute and value of res will be displayed once.

```
1 Set Integer res=0
2 do
3             --res
4             display res
5              res++
6    while(res>=0)
7      end do-while
```

Q7 — do/while with alternating increment/decrement — Step-by-step explanation

- Because res toggles in a way that makes the loop condition true every time, the loop never ends.
  - So the program runs infinitely.
  - Answer: Infinite loop (Option A).

Q8 — Nested for-loops with a break (How many times '#' is printed?) — Problem

```
1 for m = 0 to 4 step 1 do
2       for n=0 to 4 step 1 do
3               display ‘#’
4           end-for
5           if(m==2) then do
6                       break
7               end-if
8           end-for
```

Q8 — Nested for-loops with a break (How many times '#' is printed?) — Step-by-step explanation

- Outer loop m from 0..4. For m=0 inner loop prints 5 times → 5.
  - For m=1 inner loop prints 5 times → total 10.
  - For m=2 inner loop prints 5 times → total 15, then outer loop breaks.
  - So total '#' printed = 15.
  - Answer: 15 (Option A).

Q9 – Problem

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```
1 for i=0 to 4 step 1 do
2           If i==i++ + --i then do
3                       display i
4                 end-if
5           end-for
```

Q9 — Conditional true only when i == 0 — Step-by-step explanation

- Only the i==0 iteration causes the if to be true; depending on code the printed result is 0 as given on the source.
  - Answer: 0 (Option A).

Q10 – Problem

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```
1 Set Character c='7'
2 switch(c)
3            case '1': display "One"
4            case '7': display "Seven"
5          case '2': display "Two"
6        default: display "Hello"
7                        break
8       end-switch
```

Q10 — Switch with fall-through (c = 7) — Step-by-step explanation

- switch starts at case 7 and continues (falls through) because there are no breaks.
  - It executes case 7, then case 2, then default statements in order.
  - Combined printed string: 'SevenTwoHello'.
  - Answer: SevenTwoHello (Option A).

Q11 — Bitwise/shift operations and boolean check — Problem

- Given a=4, b=3, c=1 with operations like shifts and logical && final expression is a-b+c-like. What prints?

```
Integer a, b, c
Set a = 4, b = 3, c = 1
if(a >> (c-1) && b << (c + 1))
        a = a + c
Else
        b = a << c
End if
Print a - b + c
```

Q11 — Bitwise/shift operations and boolean check — Step-by-step explanation

- Evaluate (4 >> (1-1)) and (3 << (1+1)) → non-zero values treated as true.
  - If condition true → a becomes 5.
  - Print a - b + c → 5 - 3 + 1 = 3.
  - Answer: 3 (Option A).

Q12 — Simple arithmetic chain leading to p=7 — Problem

```
Integer a, p
Set a = 5
a = a + 1
a = a * 2
a = a / 2
p = a / 5 + 6
print p
```

Q12 — Simple arithmetic chain leading to p=7 — Step-by-step explanation

- Apply the steps: a increments then doubles, then halves etc.; compute p = 6/5 + 6 = 1 + 6 = 7.
  - Answer: 7 (Option D).

Q13 — XOR and conditional behavior (a=2, b=8) — Problem

```
Integer a, b, c
Set b = 8, a = 2
c = a ^ b
if(( c ^ b))
        b = 0
End if
Print b
```

Q13 — XOR and conditional behavior (a=2, b=8) — Step-by-step explanation

- a ^ b (bitwise XOR) → 2 ^ 8 = 10.
  - Then some conditional reduces b to 0 in the snippet shown on the source.
  - Source's final answer: 0 (Option B).

Q14 — Repeated additions; print a + b + c — Problem

- Starting b=40, a=20, c=20 and then repeated updates: result to be printed is a+b+c.

```
Integer a, b, c
Set b = 40, a = 20, c = 20
a = a + c
C = c + a
a = a + c
C = c + a
Print a + b + c
```

Q14 — Repeated additions; print a + b + c — Step-by-step explanation

- Follow sequence: a=20+20=40; c=20+40=60; a=40+60=100; c=60+100=160.
  - Final sum a+b+c = 100 + 40 + 160 = 300.
  - Answer: 300 (Option C).

Q15) – Problem

```
Integer a, b
Set a = 1, b = 1
a = (a ^ 1) & (1) + (b ^ 1) & (1)
Print a + b
```

Q15)— Step-by-step explanation

- Evaluate the bitwise and arithmetic pieces as shown on the source.
  - The expression simplifies to 1 according to the source explanation.
  - Answer: 1 (Option A).